



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY
DEPARTMENT
BS Computer Science
COURSE INFORMATION SHEET

Session:	Spring-2025
Course Title:	Information Security
Course Code:	CS-326T
Credit Hours:	3
Semester:	6 th
Pre-Requisite:	Computer Networks
Instructor Name:	Razia Nisar Noorani/Dr. Waleej Haider
Email and Contact Information:	rnisarn@ssuet.edu.pk/chairmancsd@ssuet.edu.pk
WhatsApp Group:	IS_Fall2025
Office Hours:	Monday & Tuesday(10:00am to 11:30am)
Mode of Teaching:	Synchronous/Asynchronous/Hybrid/Blended

COURSE OBJECTIVE:

The objective of this course is to introduce concepts related to cyber threats, cryptography and Network Security. Different security algorithms and mechanisms will be presented and solutions to security threats will be discussed.

COURSE OUTLINE:

Introduction to data and network security, goals, threats and attacks, Kill chain models, Advanced Persistent Threats, Security mechanisms, Difference between Cryptography and cryptanalysis, Traditional substitution and transposition cipher, Modern symmetric-key cryptography, Simplified DES, DES design principals, Double DES, Triple DES, Concept of Block chain with its applications, and Block cipher modes of operation, Rijndael Algorithm, Mechanism of encryption in AES, Principles of Public Key Cryptosystem, RSA Algorithm, Diffie-Hellman Key Exchange, Application of cryptographic Hash functions, Secure Hash Algorithm (SHA), Key management and distribution, Network Security Mechanisms, IPSec, Virtual Private Network, Firewalls and Intrusion Detection and Prevention Systems

COURSE LEARNING OUTCOMES (CLOs):

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
1	Explain security principles (CIA), challenges, security goals and professional practices in cybersecurity.	PLO_9 (Ethics)	C2 (Understanding)
2	Apply various algorithms and security mechanisms to provide confidentiality, integrity, and authentication	PLO_2 (Knowledge for Computing Problem)	C3 (Applying)
3	Identify appropriate techniques to analyze the problems in the discipline of network security..	PLO_3 (Problem Analysis)	C4 (Analyzing)



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RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	3 (7.4%)	3(9.5%)	4 (19%)
Assignments	3 (7.4%)	3(9.5%)	4 (19%)
Mid	15 (29.6%)	15(28.6)	-----
Final	10(55.6%)	20(52.4%)	20(62%)
Total	31 (31%)	41 (41%)	28(21%)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Quizzes	10%	10
Assignments	10%	10
Midterm Exam	30%	30
Final Exam	50%	50
TOTAL	100%	100

Recommended Book:

TEXT BOOK:

- Whitman, M.E. and Mattord, H.J. (2022). *Principles of information security*. Boston, Ma: Cengage.
- Stallings, W. (2020). *Cryptography and Network Security: Principles and Practice*. Pearson.

REFERENCE BOOKS:

- Christof P., Pelzl J., (2010), *Understanding Cryptography: A Textbook for Students and Practitioners*, Springer

LECTURE PLAN

Course Title: Information Security

Course Code: CS-326T

Week #	Week Dates	Topics	Required Reading	Key Date
1	06-03-2025 To 07-03-2025	Chapter 1: Overview of Information Security Orientation (learning objectives, performance evaluation, etc.) What is Information Security? Examples of Information Security Incidents , What is Information Security Management Human and Management Aspects of Security measure, The three concepts of Information Security (Confidentiality, Integrity, Availability) CIA Triad	Whitman Chap 1, Pg.: 1-42	



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		Basic terminologies in Information Security, Human Aspect of Information Security.		
2		Chapter 2: The Need for Security Threats and Attacks Compromises to Intellectual Property Social Engineering Information Extortion Software Attacks	Whitman Chap 2, Pg: 49-118	
3		Chapter 3: Classical Encryption Techniques Traditional substitution ciphers (Mono-alphabetic)- Additive Ceasar cipher Mono-alphabetic Ciphers Vernam Cipher Hill Cipher Playfair Cipher Book Cipher Running Key Cipher Multiplicative cipher Affine Ciphers, Playfair, Hill Cipher	Stalling, Ch.-3, Pfleeger, Ch.-2, pg. 71	
4		Traditional transposition ciphers Rail fence, simple columnar, etc	Stalling, Ch.-3, Pfleeger, Ch.-2, pg. 71	Quiz#01 Assignment 1
5		Cyber Threats and Kill Chain Methodology Describe various cyber threats and its categories ♣ Understand various threat actors ♣ Explain about motives, goals, and objectives of cyber security attacks ♣ Explain about various IoCs ♣ Understand the pyramid of pain Describe APTs ♣ Explain the cyber kill chain methodology ♣ Understand the adversary's TTPs ♣ Explain how to identify adversary behavior ♣ Understand the kill chain deep dive scenario	Internet Sources (Lecture Notes provided to Students)	
6		Chapter 4: Introduction to Block Cipher, fiestal cipher structure, Simplified DES key generation Simplified DES encryption, decryption	Stalling, Ch.-5, pg. 150	Assignment# 02
7		Data encryption standard (DES) DES design principals & Algorithm, key generation DES Encryption, Decryption	Stalling, Ch.-5, pg. 150	
8		Chapter 6: Advance Encryption Standard Rijindael Cipher, Finite Field Arithmetic, AES Structure, AES Transformation Functions, Substitute Bytes function ShiftRows Transformation, AES MixColumns functions	Stalling, Ch.-5, pg. 150	



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		Add Round Key Transformation		
9	Midterm Examination (28-04-2024 to 03-05-2025)			
10		AES Key generation and Expansion Algorithm	Stalling, Ch.-5, pg. 150	
11		Concepts of number theory , Principles of Pseudorandom Number Generation, Pseudorandom Number Generators, Linear Congruential Generators	Stalling, Ch. 4,7, pg. 103, 226	Quiz#02
12		Chapter 9: Public-key cryptography Asymmetric Encryption RSA, Blum Blum Shub Generator Stream Ciphers (RC4)	Stalling, Ch.-9, pg. 270-308	Assignment# 03
13		Chapter 10: Other Public Key System Diffie-Hellman Key Exchange	Stalling, Ch.-11, 12 pg.333-400	
14		Chapter 11: Cryptographic Hash Functions, MAC and SHA Chapter 13: Digital signatures, <u>Elgamal</u> and <u>sharmon</u> RSA and DSS approach	Stalling, Ch.-13, pg. 403	
15		User Public Key Certificate and Public Key Infrastructure (PKI) Key Certificate: Digital Signature of Public Key Public key Infrastructure (PKI) and Certificate Authority Exercise on PKI	Stalling, Chap-14, pg. 423	Quiz#03
16		Chapter 15: User Authentication Protocol Remote User Authentication Principles Kerberos	Stalling, Chap-15, Pg. 500-524	
17		State of the art work in network security (Security of IoTs), Project presentations	Internet Sources (Lecture Notes provided to Students)	
Final Examination (08-07-2025 to 19-07-2025)				

Stalling: William Stalling, **Whitman:** Michael E. Whitman, **Pfleeger:** Christof P., Pelzl